

Linear Algebra – MAT 2610

Section 4.1 (Vector Spaces and Subspaces)

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Definition

A **vector space** is a nonempty set V of objects, called *vectors*, on which there are defined two operations, called *addition* and *scalar multiplication* such that, if u, v are vectors in V and c, d are scalars, then:

- a. $u + v$ is in V (denoted as $u + v \in V$)
- b. $u + v = v + u$
- c. $(u + v) + w = u + (v + w)$
- d. There is a vector $0 \in V$ such that $u + 0 = u$
- e. For each $u \in V$, there is a vector $-u$ such that $u + (-u) = 0$
- f. $cu \in V$
- g. $c(u + v) = cu + cv$
- h. $(c + d)u = cu + du$
- i. $c(du) = (cd)u$
- j. $1u = u$



Examples

1. R^n 

2. $\{0\} \in R^n$ 

3. $\text{Span} \left\{ \begin{bmatrix} 1 \\ 2 \\ -1 \end{bmatrix}, \begin{bmatrix} -1 \\ -2 \\ 1 \end{bmatrix} \right\}$ 

4. The set of all 5×7 matrices 



Example

Let $n \geq 0$ be an integer.

Let $P_n = \{a_0 + a_1x + \cdots + a_nx^n \mid a_i \in R\}$

Note that this is the set of polynomials of degree at most n .

We can add two polynomials.

We can multiply a polynomial by a scalar.

The zero polynomial is the zero vector.

The set P_n is a vector space.

$$\begin{array}{c} 3 + x + 5x^2 + x^4 \\ 0 \\ -1 - x + x^2 - 3x^4 \end{array}$$

P_4



Definition: A set H is a **subspace** of a vector space V if it satisfies the following three properties:

- 1. The zero vector of V is in H
- 2. Whenever u and v are in H , then so is $u + v$ (closed under vector addition)
- 3. Whenever u is in H and c is a scalar, then cu is also in H (closed under scalar multiplication)



Example: The set $S = \left\{ \begin{bmatrix} s_1 \\ s_2 \\ 0 \\ 0 \end{bmatrix} \in R^4 : s_1, s_2 \in R \right\}$ is a subspace of R^4 .

Example: The set P_3 of all polynomials of at most degree 3 is a subspace of P_7



Theorem 1

If $v_1, \dots, v_p$ are in a vector space V , then $\text{Span}\{v_1, \dots, v_p\}$ is a subspace of V



Example

Example: Let $S = \left\{ \begin{bmatrix} 2a - c \\ b + c \\ 3a + b \\ a + b + c \end{bmatrix} \in \mathbb{R}^4 : a, b, c \in \mathbb{R} \right\}$ is a subspace of $\mathbb{R}^4$.

$$S = \left\{ a \begin{bmatrix} 2 \\ 0 \\ 3 \\ 1 \end{bmatrix} + b \begin{bmatrix} 0 \\ 1 \\ 1 \\ 1 \end{bmatrix} + c \begin{bmatrix} -1 \\ 1 \\ 0 \\ 1 \end{bmatrix} : a, b, c \in \mathbb{R} \right\}$$
$$= \text{Span} \left\{ \begin{bmatrix} 2 \\ 0 \\ 3 \\ 1 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 1 \\ 1 \end{bmatrix}, \begin{bmatrix} -1 \\ 1 \\ 0 \\ 1 \end{bmatrix} \right\}$$

By Thm 1, S is a subspace of $\mathbb{R}^4$



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